QUASI-DM-NESS TESTED THROUGH LOCALLY QUASI-FINITE COVERS OVER TEST OBJECTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The characterization of local quasi-finiteness through covers over every test object. Invoked inside the counted proof (source lines 5104-5112). Supplies the covers used to test the condition.
- The criterion for checking a property of a morphism along a flat finitely presented cover. Invoked inside the counted proof (source lines 5113-5125). Transfers the property between the morphism and its base change.
- The relation between quasi-DM-ness and local quasi-finiteness of the comparison. Used at the end of the counted proof (source lines 5126-5131). Closes the equivalence.

Source reference. Original lemma lemma-base-change-separated: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 703–703. Navigation pointer only: the counted text refers to the Stacks item lemma-base-change-separated, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-check-separated-covering: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 727–727. Navigation pointer only: the counted text refers to the Stacks item lemma-check-separated-covering, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Date: stacks-project:stacks-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma lemma-properties-covering-imply-diagonal: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 1117–1117. Navigation pointer only: the counted text refers to the Stacks item lemma-properties-covering-imply-diagonal, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-separated-implies-morphism-separated: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 1061–1061. Navigation pointer only: the counted text refers to the Stacks item lemma-separated-implies-morphism-separated, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-separated-over-separated: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 888–888. Navigation pointer only: the counted text refers to the Stacks item lemma-separated-over-separated, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-trivial-implications: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 682–682. Navigation pointer only: the counted text refers to the Stacks item lemma-trivial-implications, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original theorem theorem-quasi-DM: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4286–4286. Navigation pointer only: the counted text refers to the Stacks item theorem-quasi-DM, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks. The following are equivalent*

1. *f is quasi-DM,*
2. *for any morphism $V \to \mathcal{Y}$ with V an algebraic space there exists a surjective, flat, locally finitely presented, locally quasi-finite morphism $U \to \mathcal{X} \times_{\mathcal{Y}} V$ where U is an algebraic space, and*
3. *there exist algebraic spaces U, V and a morphism $V \to \mathcal{Y}$ which is surjective, flat, and locally of finite presentation, and a morphism $U \to \mathcal{X} \times_{\mathcal{Y}} V$ which is surjective, flat, locally of finite presentation, and locally quasi-finite.*

Proof. The implication (2) $\Rightarrow$ (3) is immediate.

Assume (1) and let $V \to \mathcal{Y}$ be as in (2). Then $\mathcal{X} \times_{\mathcal{Y}} V \to V$ is quasi-DM, see Lemma lemma-base-change-separated. By Lemma lemma-trivial-implications the algebraic space V is DM, hence quasi-DM. Thus $\mathcal{X} \times_{\mathcal{Y}} V$ is quasi-DM by Lemma lemma-separated-over-separated. Hence we may apply Theorem theorem-quasi-DM to get the morphism $U \to \mathcal{X} \times_{\mathcal{Y}} V$ as in (2).

Assume (3). Let $V \to \mathcal{Y}$ and $U \to \mathcal{X} \times_{\mathcal{Y}} V$ be as in (3). To prove that f is quasi-DM it suffices to show that $\mathcal{X} \times_{\mathcal{Y}} V \to V$ is quasi-DM, see Lemma lemma-check-separated-covering. By Lemma lemma-properties-covering-imply-diagonal we see

that $\mathcal{X} \times_{\mathcal{Y}} V$ is quasi-DM. Hence $\mathcal{X} \times_{\mathcal{Y}} V \to V$ is quasi-DM by Lemma lemma-separated-implies-morphism-separated and (1) holds. This finishes the proof of the lemma. □